Arslan Khalid

Python Developer

iarslankhalid@yahoo.com | LinkedIn | GitHub | WhatsApp | (+92) 341-9507071

Education

B.Sc. Electrical Engineering
University of Gujrat
CGPA: 3.12
Nov 2020 – Jul 2024

Relevant Courses:
- Advanced ML Algorithms by Stanford University
- Supervised Machine Learning by Stanford University
- Django Web Framework by Meta

Skills

Languages:	Python · C++ · TypeScript · JavaScript · SQL · HTML/CSS
Frameworks/Libraries:	FastAPI · Django · React · Next.js · Node.js · LiveKit · OpenCV · TensorFlow · PyTorch
Databases & Tools:	PostgreSQL · Redis · Prisma · Docker · WebSockets · Git · GitHub
AI & Machine Learning:	LLMs (OpenAI, Claude) · RAG · AI Agents · Prompt Engineering · Audio Synthesis · Computer Vision
Soft Skills:	Leadership · Team Coordination · Time Management · Critical Thinking · Problem Solving

Experience

- **Zylo Solutions (Remote)** — Wyoming, USA
  AI Development Team Lead — *Nov 2025 – Present*
  - Lead cross-functional teams of 6–10 engineers delivering three production AI platforms (ZARM, Scale Connect, AI-FSL), owning architecture, delivery timelines, and engineering standards.
  - ZARM (team of 10): built an AI consulting platform that compresses weeks of manual discovery into a structured 5-day sprint — AI agents generate interview questions, extract pain points and risks from meetings, and score business processes on AI readiness, delivering clients a prioritized AI roadmap in one week.
  - Scale Connect (team of 8): delivered a property portfolio & financial dashboard centralizing real-estate metrics, with structured LLM prompting for automated, data-driven financial strategy insights.
  - AI-FSL (team of 6): delivered a multi-tenant French-language education platform adopted by Canadian school boards — AI lesson-plan generation, CEFR-aligned proficiency assessment with speech-to-text, and curriculum-aware AI chat, cutting lesson preparation from hours to minutes.
  - Established CI/CD, testing, and LLM prompt/model evaluation pipelines across all project teams.

- **Zylo Solutions (Remote)** — Wyoming, USA
  Machine Learning Engineer — *May 2025 – Nov 2025*
  - Served as AI Solution Architect for Brewly, an AI cafe management system: weather- and event-aware inventory forecasting with 90%+ accuracy, time-of-day smart bundles that boosted 3–5pm sales by up to 30%, and a LiveKit-powered AI drive-thru agent with sub-700ms response latency — deployed at Taken Cafe, Dubai.
  - Built Flossie.ai, a voice front-desk assistant for dental clinics (Vapi + Twilio) that answers calls, books appointments, sends doctor reminders, and supports multi-branch practices — live in 10+ clinics.
  - Developed an AI meeting notetaker that joins Google Meet, Zoom, and Teams calls and delivers transcripts, summaries, and actionable next steps.
  - Built EasyGen, a LinkedIn post-generation platform trained on top-influencer content, shipped with a companion Chrome extension.

- **GulzarSoft (On-site)** — Gujrat, Pakistan
  Machine Learning Engineer — *Nov 2024 – Apr 2025*
  - Designed and implemented a collaborative filtering model for a car recommendation system, enhancing recommendation accuracy by 30%.

- **Coders Launchpad (Remote)** — Dubai, UAE
  Machine Learning Engineer — *Dec 2023 – Apr 2024*

- Developed a voice-to-voice conversion model for the Quran Project using RVC technology to transform users' voices while maintaining high-quality audio synthesis.
- Conducted rigorous training and optimization of the voice modulation models to ensure high accuracy, pronunciation clarity, and natural-sounding synthesis.

- **CodeClause (Remote)** Remote
  Artificial Intelligence Intern *Feb 2024 – Mar 2024*
  - Developed a real-time speech recognition and speech-to-text conversion application to transcribe audio streams.
  - Implemented a personality prediction system leveraging computer vision models to analyze facial expressions and cues.

- **Freelancer** Gujrat, Pakistan
  Machine Learning Engineer *May 2021 – Mar 2024*
  - Designed and deployed custom machine learning models, automation scripts, and backend integrations for global clients.
  - Built robust web scraping data pipelines and automated notification workflows using Python, Celery, and external APIs.

- **Intelligent Systems Lab (On-site)** Gujrat, Pakistan
  Machine Learning Intern *Sep 2023 – Oct 2023*
  - Developed a machine learning model for depth estimation and path planning for an autonomous quadruped robot.
  - Integrated stereo vision cameras and various onboard sensors to enable obstacle detection and real-time navigation.

Projects

- **Jarvis — Real-Time AI Voice Assistant**
  LiveKit, Python, WebSockets, Wake-Word Detection
  - Architected a persistent, low-latency AI voice assistant leveraging LiveKit real-time infrastructure.
  - Trained and deployed custom wake-word detection models, optimizing streaming connection latency to achieve near-instantaneous conversational responsiveness.
  - Bridged continuous audio streaming with backend AI processing pipelines, enabling seamless and natural user interactions.

- **Project Syndra — Social Media Automation Engine**
  Python, Node.js, Docker, Web Scraping APIs
  - Developed an end-to-end automated news aggregation and publishing engine tailored for social media platforms using Python and Node.js.
  - Built a robust data pipeline to autonomously scrape, dynamically rank, and filter visual news content based on custom relevance algorithms.
  - Integrated external APIs to schedule and publish high-quality visual content, completely automating the content pipeline and saving 20 hours of manual curation weekly.

- **Cortex — LLM Reasoning Visualization Tool**
  React, TypeScript, Python, FastAPI, WebSockets
  - Designed and deployed a full-stack software system dedicated to tracking, logging, and visually mapping the internal reasoning chains of Large Language Models (LLMs).
  - Built a highly responsive frontend dashboard and robust backend architecture to process and display complex AI inference steps in real-time.
  - Enabled developers to effectively debug, audit, and optimize LLM prompts and agent behaviors through comprehensive visual analytics.

Awards

- Achieved **top 5** position in Devathon, a hackathon organized by Devsinc, Pakistan.
- Earned the **Master** title on Kaggle for excellence in data science and machine learning competitions.
- Received funding for my Final Year Project (FYP) through the **IGNITE NIGRI** program.
- Secured **first position** in 9th, 10th, 11th, and 12th grades for academic excellence.
- Won the **typing competition** organized by 10FastFingers.